

Abdellah Fourtassi

Associate Professor (Maître de Conférences HDR) in Computer Science

Aix-Marseille Université · LIS / TALEP · Institute of Language, Communication & the Brain (ILCB)
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RESEARCH FOCUS I build machine-learning models that learn from multimodal interaction under cognitively plausible constraints. The goal is to develop and test formal theories of how interaction shapes language learning and communication in humans and machines.

ACADEMIC APPOINTMENTS & ROLES

Aix-Marseille Université - Associate Professor (Maître de Conférences HDR), Computer Science	2025-present
Aix-Marseille Université - Assistant Professor (Maître de Conférences), Computer Science	2019-2025
Director, Computation in Communicative Development (CoCoDev) team	2019-present
Deputy Director for Research, Institute of Language, Communication & the Brain (ILCB)	2024-present
Co-lead, <i>NLP & Cognition</i> national working group, GDR-CNRS	2025-present
Co-Director, Master in Cognitive Science (MASCO), Aix-Marseille Université	2020-2021
Associate Editor, <i>Dialogue & Discourse</i>	2026-present
Visiting Researcher, INRIA Paris (Délégation Recherche)	2021-2022
Postdoctoral Research Fellow, Department of Psychology, Stanford University	2016-2019

EDUCATION

Ph.D. in Cognitive Science, École Normale Supérieure (ENS), Paris - summa cum laude	2015
Master's in Cognitive Science (CogMaster), EHESS, Paris	2012
Science & Engineering Diploma, École Polytechnique, Paris	2012
CPGE/MP*, Mathematics & Physics, Lycée Moulay Youssef, Rabat	2006-2008

RESEARCH FUNDING

ANR JCJC - Principal Investigator, €287k	2022-2025
<i>Mastering the Art of Conversation in Middle Childhood (MACoMiC)</i>	

SELECTED HONORS & DISTINCTIONS

- Keynote speaker: SIGDIAL (2025), LM Playschool Workshop at EMNLP (2026), Child Language Symposium (2026)
- Aix-Marseille Université Best Ph.D. Award (2024), awarded to M. Nikolaus; co-supervised with L. Prévot
- Marie Skłodowska-Curie European Fellowship, €185k (2019; declined after accepting tenured position)
- Stanford Data Science Seed Grant, \$50k (2018)
- Fyssen Foundation Postdoctoral Study Grant, €40k (2016)
- France-Stanford Center Collaborative Project Award, \$15k (2015-2016)
- CoNLL Best Paper Award finalist (for Fourtassi & Dupoux, 2014)
- ACL Best Student Paper Award, \$1,000 (for Fourtassi & Dupoux, 2013)

PUBLICATIONS

Book

Fourtassi, A. (forthcoming). *Neural Networks for a Theory of Language Development*. Cambridge University Press.

Edited collections

- Agee, L., Monfils, M., & Fourtassi, A. (2025). Editorial: The Neural and Behavioral Methods of Social Learning. *Frontiers in Human Neuroscience*.
- Apidianaki, M., Fourtassi, A., & Padó, S. (2024). Editorial: Language Learning, Representation, and Processing in Humans and Machines. *Computational Linguistics*.

Journal articles

1. Brahimi, Y., & Fourtassi, A. (under review). Slow but Predictive: Turn-Taking Development in Child-Adult Conversation.
2. Agrawal, A., Favre, B., & Fourtassi, A. (in press). Scaffolding Early Dialogue: A Unified Account of Response Contingency in Child-Caregiver Interaction. *Cognitive Science*.
3. Nikolaus, M., & Fourtassi, A. (2026). Modeling Children's Grammar Learning via Caregiver Feedback in Natural Conversations. *Philosophical Transactions of the Royal Society B*, 381.
4. Gupta, S., Mazzocconi, C., & Fourtassi, A. (2025). Toward a child-centered, interactive approach to multimodal language development. *First Language*, 45(6).
5. Mazzocconi, C., O'Brien, B., Bodur, K., & Fourtassi, A. (2025). Do Children Laugh Like Their Parents? Conversational Laughter Mimicry and Acoustic Alignment in Middle Childhood. *Journal of Nonverbal Behavior*, 49.

6. Bodur, K., Fourtassi, A., Fredouille, C., & Meunier, C. (2025). Speech Reduction in Middle Childhood: The Case of Child-Caregiver Interactions. *First Language*.
7. Agee, L., Fourtassi, A., & Monfils, M. (2025). Social Context as a Source of Variability in the Psychological Sciences. *Frontiers in Behavioral Neuroscience*, 18.
8. Cabiddu, F., Nikolaus, M., & Fourtassi, A. (2025). Comparing Children and Large Language Models in Word Sense Disambiguation: Insights and Challenges. *Language Development Research*, 5(1).
9. Bodur, K., Nikolaus, M., Prévot, L., & Fourtassi, A. (2023). Using Video Calls to Study Children's Conversational Development: The Case of Backchannel Signaling. *Frontiers in Computer Science*, 5.
10. Wilson, K., Frank, M. C., & Fourtassi, A. (2023). Conceptual Hierarchy in Child-Directed Speech: Implicit Cues Are More Reliable. *Journal of Cognition & Development*, 24(4).
11. Nikolaus, M., & Fourtassi, A. (2023). Communicative Feedback in Language Acquisition. *New Ideas in Psychology*, 68, 100985.
12. Nikolaus, M., Maes, E., Auguste, J., Prévot, L., & Fourtassi, A. (2022). Large-Scale Study of Speech Acts' Development in Early Childhood. *Language Development Research*, 2(1).
13. Jiang, H., Frank, M. C., Kulkarni, V., & Fourtassi, A. (2022). Stability and Change in Caregivers' Word Usage Across Early Childhood. *Cognitive Science*, 46(7).
14. Fourtassi, A., Regan, S., & Frank, M. C. (2021). Continuous Developmental Change Explains Discontinuities in Word Learning. *Developmental Science*, 24(2).
15. Fourtassi, A., & Frank, M. C. (2020). How Optimal Is Word Identification Under Multimodal Uncertainty? *Cognition*, 199.
16. Fourtassi, A., Bian, Y., & Frank, M. C. (2020). The Growth of Children's Semantic and Phonological Networks: Insights from 10 Languages. *Cognitive Science*, 44(7).

Peer-reviewed conference proceedings

(Some publication titles are abbreviated for space)

1. Jumelet, J., Fourtassi, A., ... & Choshen, L. (2026). BabyBabelLM: Multilingual Benchmark of Developmentally Plausible Training. **EACL**.
2. Brahimi, Y., Blanc, C., & Fourtassi, A. (2026). Predicting Turn-Taking in Child-Adult Conversations Using Voice Activity Projection. **IWSDS**.
3. Liu, J., & Fourtassi, A. (2025). Benchmarking LLMs for Mimicking Child-Caregiver Language in Interaction. **SemDial**.
4. Agrawal, A., Favre, B., & Fourtassi, A. (2025). Identifying Repair Opportunities in Child-Caregiver Interactions. **SemDial**.
5. Agrawal, A., Nikolaus, M., Favre, B., & Fourtassi, A. (2024). Contingency in Child-Caregiver Conversations. **LREC-COLING**.
6. Nikolaus, M., Agrawal, A. ... & Fourtassi, A. (2024). Grammaticality in Child-Caregiver Conversations. **LREC-COLING**.
7. Goumri, D., Agrawal, A. ... & Fourtassi, A. (2024). A Corpus of Child-Caregiver Face-to-Face Conversations. **LREC-COLING**.
8. Schweitzer, M., Montagnini, A., Fourtassi, A., & Schatz, T. (2024). Audio-Visual Speech Recognition. **NeurIPS Behavioral ML Workshop**.
9. Goumri, D., Becerra, L., & Fourtassi, A. (2024). Child-Caregiver Gaze Dynamics in Naturalistic Face-to-Face Conversations. **CogSci**.
10. Agrawal, A., Favre, B., & Fourtassi, A. (2024). Analyzing Communicative Intent Coordination in Child-Caregiver Interactions. **CogSci**.
11. Peirolo, M., Xu, Z., & Fourtassi, A. (2024). Development of Flexible Role-Taking in Conversations Across Preschool. **CogSci**.
12. Goumri, D., Janssoone, T., Becerra, L., & Fourtassi, A. (2023). Detection of Gaze and Smile in Children's Video Calls. **ICMI Companion**.
13. Cabiddu, F., Nikolaus, M., & Fourtassi, A. (2023). Modeling Children's Word Sense Disambiguation with LLMs. **CogSci**.
14. Nikolaus, M., Prévot, L., & Fourtassi, A. (2023). Communicative Feedback in Response to Children's Grammatical Errors. **CogSci**.
15. Mazzocconi, C., O'Brien, B., ... & Fourtassi, A. (2023). Mimicking Laughter in Child-Caregiver Conversation. **CogSci**.
16. Agrawal, A., Liu, J., ... & Fourtassi, A. (2023). Development of Multimodal Turn Coordination in Conversations. **CogSci**.
17. Tiran, T., Meewis, F., Fourtassi, A., & Dautriche, I. (2023). Typology of Topological Relations Using Machine Translation. **CogSci**.
18. Nikolaus, M., Salin, E., Ayache, S., Fourtassi, A., & Favre, B. (2022). Do Vision-Language Models Learn Predicate-Nouns? **EMNLP**.
19. Liu, J., Nikolaus, M., Bodur, K., & Fourtassi, A. (2022). Predicting Backchannel in Child-Caregiver Conversation. **ICMI Companion**.
20. Misiek, T., & Fourtassi, A. (2022). Caregivers Exaggerate Their Lexical Alignment to Young Children Across Several Cultures. **SemDial**.
21. Hallart, C., Peirolo, M., Xu, Z., & Fourtassi, A. (2022). Development of Contingency in Early Child-Caregiver Interaction. **SemDial**.
22. Bodur, K., Nikolaus, M., Prévot, L., & Fourtassi, A. (2022). Backchannel Behavior in Child-Caregiver Video Calls. **CogSci**.
23. Nikolaus, M., Prévot, L., & Fourtassi, A. (2022). Communicative Feedback Supports Intelligible Speech in Early Childhood. **CogSci**.
24. Nikolaus, M., & Fourtassi, A. (2021). Interaction Between Perception- and Production-Based Learning in Early Development. **CoNLL**.
25. Dary, F., Nasr, A., & Fourtassi, A. (2021). TALEP at CMCL 2021: Predicting Eye-Tracking Data Challenge (2nd/30). **CMCL**.
26. Bodur, K., Nikolaus, M., ... & Fourtassi, A. (2021). ChiCo: A Corpus for the Study of Child Conversation. **ICMI Companion**.
27. Nikolaus, M., & Fourtassi, A. (2021). Acquisition of Semantic Knowledge from Cross-Situational Learning in Neural Networks. **CMCL**.
28. Nikolaus, M., Maes, E., Auguste, J., Prévot, L., & Fourtassi, A. (2021). Large-Scale Study of Speech-Act Development. **CogSci**.
29. Nikolaus, M., Maes, E., & Fourtassi, A. (2021). Modeling Speech-Act Development in Early Childhood. **CogSci**.
30. Dary, F., Fourtassi, A., & Nasr, A. (2021). Low-Level Linguistic Tasks for Reading-Time Prediction. **CogSci**.
31. Misiek, T., Favre, B., & Fourtassi, A. (2020). Development of Multi-Level Linguistic Alignment in Child-Adult Conversations. **CMCL**.
32. Fourtassi, A. (2020). Word Co-Occurrence in Child-Directed Speech Predicts Children's Free Word Associations. **CMCL**.
33. Fourtassi, A., Wilson, K., & Frank, M. C. (2020). Discovering Conceptual Hierarchy Through Explicit and Implicit Cues. **CogSci**.
34. Fourtassi, A., Scheinfeld, I., & Frank, M. C. (2019). Development of Abstract Concepts in Children's Early Lexical Networks. **CMCL**.

35. Fourtassi, A., Regan, S., & Frank, M. C. (2019). Continuous Developmental Change and Discontinuities in Word Learning. **CogSci**.
36. Fourtassi, A., & Dupoux, E. (2019). Phoneme Learning and Taxonomic Similarity of Semantic Referents. **CogSci**.
37. Fourtassi, A., Bian, Y., & Frank, M. C. (2018). Word Learning as Network Growth: A Cross-Linguistic Analysis. **CogSci**.
38. Fourtassi, A., & Frank, M. C. (2017). Word Identification Under Multimodal Uncertainty. **CogSci**.
39. Fourtassi, A., & Dupoux, E. (2016). The Role of Word-Word Co-Occurrence in Word-Meaning Learning. **CogSci**.
40. Fourtassi, A., & Dupoux, E. (2014). Rudimentary Lexicon and Semantics Help Bootstrap Phoneme Acquisition. **CoNLL**.
41. Fourtassi, A., Schatz, T., Varadarajan, B., & Dupoux, E. (2014). Bottom-Up and Top-Down Information in Phoneme Learning. **ACL**.
42. Fourtassi, A., Dunbar, E., & Dupoux, E. (2014). Self-Consistency as an Inductive Bias in Early Language Acquisition. **CogSci**.
43. Fourtassi, A., Borschinger, B., Johnson, M., & Dupoux, E. (2013). Why Is English So Easy to Segment? **CMCL**.
44. Fourtassi, A., & Dupoux, E. (2013). A Corpus-Based Evaluation Method for Distributional Semantic Models. **ACL-SRW**.
45. Jansen, A., et al. (2013). Zero-Resource Speech Technologies and Models of Early Language Acquisition. **ICASSP**.

INVITED TALKS

Keynotes

- LM Playschool Workshop at EMNLP (Budapest, 2026)
- Child Language Symposium (London, 2026)
- SIGDIAL (Avignon, 2025)

Selected invited lectures and expert meetings

- University of Vienna, Current topics in AI, guest lecture (2026)
- International Congress of Infant Studies (ICIS, online event, 2025)
- Mila Quebec AI Institute workshop on NLP & Cognitive Science (2024)
- UCL, Psychology and Language Sciences (2023, 2024)
- Dagstuhl Seminar: Developmental Machine Learning (2022)
- OECD workshop: AI and the Productivity of Science (2021)

MENTORSHIP

Ph.D supervision - primary advisor

- Marianne Schweitzer (2025-present), co-advised with C. François
- Youcef Brahimi (2025-present)
- Dhia Elhak Goumri (2022-2026), co-advised with L. Becerra; defended March 2026
- Abhishek Agrawal (2022-2025), co-advised with B. Favre; defended October 2025
- Mitja Nikolaus (2020-2023), co-advised with L. Prévot; defended October 2023; *AMU Best Ph.D. Award 2024*

Ph.D supervision - secondary advisor

- Mathias Vigouroux (2025-present), Jing Liu (2024-present), Franck Dary (2018-2022)

Postdoctoral supervision

- Giulia Rambelli (2026-present), Monica Barbir (2024-present), Chiara Mazzocconi (2020-2023)

Additional supervision

- Supervised 14 Master's interns, 17 summer interns; multiple projects resulted in peer-reviewed publications
- Member of 13 thesis advisory committees

TEACHING

- Building Minds: The Programming Tools - Master in Cognitive Science (2025-present)
- Machine Learning for Cognitive Science - MASCO / Master in Computer Science (2024-present; with T. Schatz)
- Data Science (L1 Computer Science, 2025-present)
- Introduction to Machine Learning (L3 Computer Science, 2022-2023)
- Introduction to Natural Language Processing (M1 Computer Science, 2020-2023)
- Probability Theory for Computer Science (M1, 2022)
- Theories in Cognitive Science (MASCO, 2020-2022)
- ILCB Summer School: week-long course on deep learning and language development (2025, 2026)
- ETAL National Summer School on NLP: When NLP Meets Cognitive Science (2023)

EDITORIAL, REVIEWING & SERVICE

Editorial and conference roles

- Associate Editor, *Dialogue & Discourse* (2026-present)
- Area Chair, ACL Rolling Review (2025-present)

- Program Chair, ACL Workshop on Computational Developmental Linguistics (2026)
- Meta-Reviewer, CogSci (2026)
- Senior Area Chair, COLING (2025)
- Associate Editor, *Frontiers in Cognitive Neuroscience* (2023-2025)
- Guest Editor, *Computational Linguistics* (2024) and *Frontiers in Human Neuroscience* (2025)

Reviewing

- Funding agencies: ANR, NSF, British Academy, Royal Society, JST
- Selected journals: Psychological Review, Trends in Cognitive Sciences, Proceedings B, Cognition, Cognitive Science, Child Development, Developmental Psychology, JEP: General, Behavior Research Methods
- Selected conferences: ACL Rolling Review, CogSci, ICMI, CoNLL, CDS, ACII, CMCL

Service & Events (recent)

- Co-organizer, national workshop on *Developmental AI*, GDR-CNRS, 2026 (with M. Fort, T. Hueber)
- Evaluation panel, France-Japan bilateral funding, ANR-JST CREST, 2026
- Co-organizer, CIVIS Europe-wide collaborative workshop, 2026 (with M. Jover)
- Co-organizer, national workshop on *NLP & Cognition*, GDR-CNRS, 2025 (with P. Denis, L. Prévot)
- Selection panel, Associate Professor position in Computer Science, Université de Lille, 2025

PUBLIC OUTREACH

- Bringing the lab to the classroom, ANR-funded Festival Explore (2026)
- Interview: AI and Cognitive Science, Photizo podcast (2025)
- Interactive MACoMiC project stand, Nuit des chercheurs (2023)
- Public outreach article in *The Conversation* on infants learning conversation (2022)